

Instructor: Date:

Learner: Regional Branch:

Program: Center:

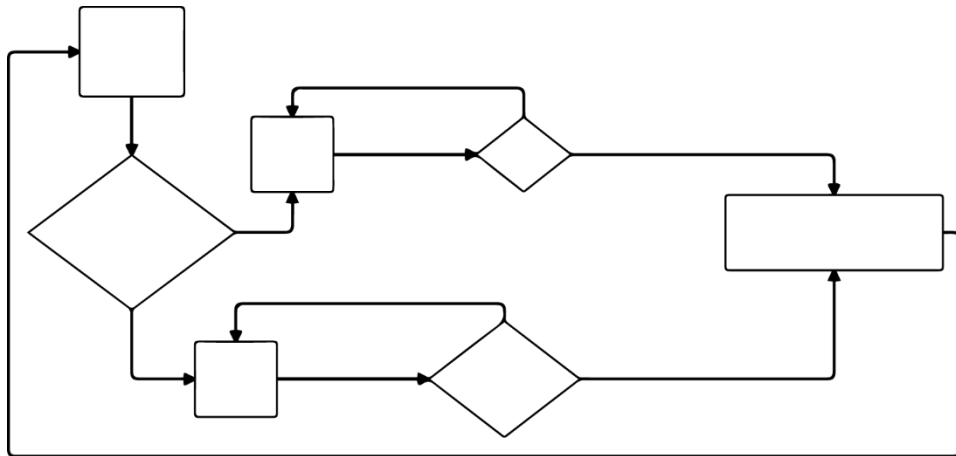
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

Essential Elements in Programming



Learning Outcome:

Talk about principles, concepts, and characteristics of general-purpose computer programming language in spoken and written situations.

Ask your partner:

- What fundamental programming elements do you think are needed in order to start programming?
- Do you think any person can learn how to program? Why? Why not?

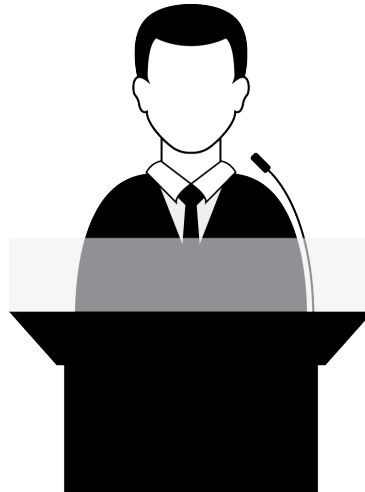
Can Do Statements:

- I can list the five basic elements of programming.
- I can give some characteristics of the five basic elements of programming.
- I can identify elements in a flow chart about the five basic elements of programming.



Let's Explore!

A Lecture on the Basics of Programming

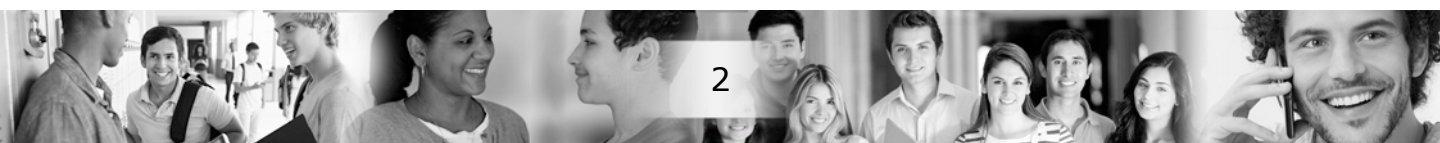


Instructions: Read the text below.

Good morning, everybody. Before we talk about common programming languages, I would rather start by talking about the five basic elements that have made programming possible. To start with, you always need to have an input. The input is the information that enters into the system and is needed for a program to work. Programmers have used keyboards, text files, and touchscreens as ways of entering data into a program. The next element is arithmetic. This has always been the most difficult one as it is where all the mathematical operations happen. Also, if the program doesn't directly need operations, it might still need them for the programming itself.

The next step is the output. Programming itself has been designed for the purpose of having an outcome. After the processing of the input has been done, the sole purpose of a program is to give a result. This way, the outcome might be the most tangible part of programming.

When deciding what kind of looping you would rather use in order to repeat an operation, a number in the program is an essential part of programming. Would you rather use the event-controlled loop or the counter-controlled loop? You have to check which is best in order for it to suit your needs. Do you need this eternal loop to be stopped by an external event or by a predetermined counter? Always ask yourself what is best for the program. Don't forget that an important element for the loop is the conditional, which determines when the loop finishes. Finally, people have always needed maps to guide themselves through processes. In this case, you need a flowchart to make sure you are following the right steps in a process.



**Let's Practice!**

The The speech was about the five basic elements of programming.

Instructions: Select the correct word or phrase to complete each sentence.

1. I would rather start by talking about the five basic elements that have (make / made / makes) programming possible.
2. Keyboards, text files or a touchscreen have (been / were / are) common ways of entering data into a program.
3. Arithmetic (have always been / has been always / has always been) the most difficult part of programming.
4. Programming itself has (been designing / been designed / been design) for the purpose of having an outcome.
5. After the processing of the input (has been done / have been done / have been doing), the sole purpose of a program is to give a result.

**Let's Practice!**

Identifying the Basics of Programming

Instructions: Match column A with column B.

COLUMN A

1. You would rather stop the repetition
2. Would you rather use the
3. Would you rather have this eternal loop stopped
4. The outcome might be the
5. Deciding what kind of looping you would rather use

COLUMN B

- a. most tangible part of programming.
- b. in order to repeat an operation, a number in the program is an essential part of programming.
- c. of an operation in the program.
- d. event-controlled loop or the counter-controlled loop?
- e. by an external event or by a predetermined counter?





Extension Activity

Getting Deep on the Basic Elements

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information about basic elements on programming.
- Write down some notes and design a chart which you show the main features of each basic element.
- Share with your partner and answer all his questions.

Self-Assessment

Yes No

- | | | |
|--|-----------------------|-----------------------|
| 1. I look for information about basic elements on programming. | ☐ | ☐ |
| 2. I design a chart to show the main features of each basic element. | ☐ | ☐ |
| 3. I talk about basic elements on programming. | ☐ | ☐ |



Target Words

- **looping** (noun): a set of instructions that is repeated a determined number of times
e.g. Deciding what kind of looping you would rather use in order to repeat an operation a certain number of times in the program is an essential part of programming.
- **flowchart** (noun): a diagram of the steps in a process
e.g. In this case, you need a flowchart to make sure you are following the right steps in a process.
- **event-controlled loop** (noun): a part of a program that determines when the loop ends by introducing an external event
e.g. Would you rather use the event-controlled loop or the counter-controlled loop?
- **input** (noun): any kind of data that is entered into a program
e.g. The input is the information that enters into the system and is needed for a program to work.
- **arithmetic** (noun): the method or way mathematical operations work in a program
e.g. The next element is arithmetic; this has always been the most difficult one, as it is where all the mathematical operations happen.
- **conditional** (noun): the event that determines when the coding loop stops
e.g. Don't forget that an important element for the loop is the conditional, which determines when the loop finishes.



Learning Points

Present Perfect: Affirmative

You use the **present perfect** to talk about an action that started in the past and is still happening now.

You can also use the **present perfect** to talk about an action that is still happening or that stopped recently, but somehow it has an effect on the present. The **present perfect** emphasizes the result of an action as well.

The structure of the **present perfect** in **affirmative** statements is as follows:

Subject + have/has + verb in past participle

Examples:

*Since I started programming, I **have** always **taken** into account the input and output. She **has** just **finished** writing the event-controlled loop.*

Examples

- *Programmers have used keyboards, text files, and touchscreens as ways of entering data into a program.*
- *I would rather start by talking about the five basic elements that have made programming possible.*



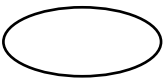
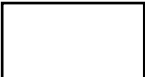
Learning Points

Flowcharts to Present Basic Elements of Programming

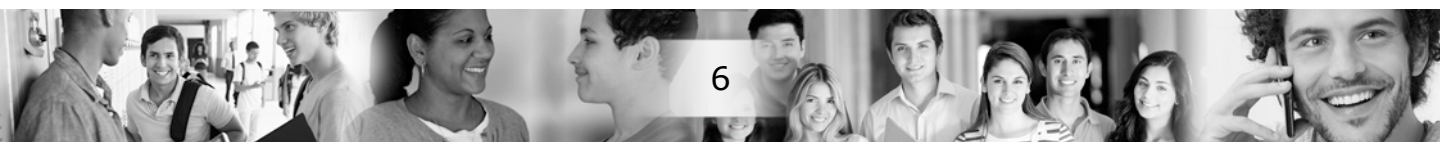
A **flowchart** is a drawing that represents the different stages of a process.

You use a **flowchart** in programming to diagrammatically represent an algorithm. A workflow shows the steps of a process, which are connected by arrows in order to indicate the sequence of the elements.

You need to use specific symbols to make a flowchart.

	Terminal It is used at the beginning and at the end of the flowchart.
	Input/Output It is used to represent input and output operations.
	Arrow It indicates the flow of logic.
	Decision It is used when there are two options: true and false.
	Rectangle This symbol is used for arithmetic operations.

Let's check an example of a flowchart that represents the steps of the process of an ATM machine.





Learning Points

Examples

